Cyrus Guided Notes: MIT 6.241J Dynamic Systems and Control

MIT course pack

2026-05-15

Course source

Official source: `https://ocw.mit.edu/courses/6-241j-dynamic-systems-and-control-spring-2011/`

Why this course is in the system. Core control course for state space, controllability, observability, stability, and feedback.

Learning output. Priority course for a complete guided PDF connected to existing control formula visuals.

Prerequisite checkpoint

- Linear algebra
- Eigenvalues
- Differential equations

Formula checkpoint

Do not memorize these first. Read each formula as a sentence: what changes, what is observed, what is optimized, and what output it produces.

$$\dot{\boldsymbol{x}} = A\boldsymbol{x} + B\boldsymbol{u}$$
$$e^{At} = \mathcal{L}^{-1}\{(sI - A)^{-1}\}$$
$$\mathrm{rank}[B\ AB\ \cdots\ A^{n-1}B] = n$$
$$\boldsymbol{u} = -K\boldsymbol{x}$$

Visual page

Make state propagation visible: A moves the state, B injects input, K closes feedback.

GoodNotes pages

- Page MIT-C001: state space

- Page MIT-C002: controllability
- Page MIT-C003: feedback

Web interaction

A two-state system where dragging eigenvalues changes stability.

Obsidian and Notion

Layer	Output
Obsidian	MIT Control -> 6.241J dynamic systems
Notion	Control gate status and solved derivation evidence.

First study move

Open the official source, copy the prerequisite checkpoint into GoodNotes, then write one formula in your own words before watching or reading more.